

SAPNA TIWARI

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EDUCATION

Vellore Institute of Technology (VIT), Bhopal

2023 – 2027

B.Tech, Computer Science & Engineering | CGPA: 8.98

SKILLS

Languages: Java, JavaScript, React.js, HTML

Frameworks/Technologies: RESTful APIs, Node.js, Vite, MongoDB, ExpressJS, Socket.io, WebRTC, bcrypt.js, JWT

AI/ML: LangChain, RAG Pipelines, FAISS, Prompt Engineering, Google Gemini API

Core CS: Data Structures & Algorithms (150+ problems solved), Computer Networks, DBMS, OS

PROJECTS

YouTube RAG Assistant [GitHub](#) | [Live Demo](#)

FastAPI, LangChain, FAISS, HuggingFace, RAGAS, JavaScript, HTML

- Developed a Chrome extension + FastAPI backend that answers questions over YouTube video transcripts via RAG (transcript chunking → FAISS vector search → Qwen 2.5 7B via HuggingFace Inference API)
- Implemented manual evaluation harness against a hand-labeled 100-question test set, measuring Precision@2 = 70% (k=2), then cross-validated using RAGAS (context precision 79.2%, faithfulness 81.1%, answer relevancy 78.5%).
- Cut repeated-query latency 85% (20s → 3s) by caching per-video FAISS indexes, keyed by video ID; tested across 100 videos (4,832 transcript chunks, chunk_size=1000, overlap=200).

AI Habit Tracker [GitHub](#)

MERN (MongoDB, Express, React, Node.js), Google Gemini 2.5 Flash, PWA, VAPID, JWT, bcrypt

- Engineered a progressive web application for habit tracking using Node.js and MongoDB Atlas, calculating streaks with date-fns logic for over 500 active users.
- Deployed Google Gemini 2.5 Flash into five user-facing AI capabilities, achieving sub-two-second response times by leveraging real user habit and streak data for prompt engineering.
- Built an idempotent streak-reminder cron job (node-cron, VAPID Web Push) with per-user-per-day state checks, validated against 1000 simulated users with zero duplicate notifications.

ACTIVITIES

VIT Bhopal Cyber Security Club — Event Management, Cyber Carnival '26 *AdVITya'26*

- Coordinated event outreach for Cyber Carnival '26, helping draw 1,000+ student footfall.

ICDCC 2024 — Website Development Team Member

Nov – Dec 2024

- Built the homepage and speaker section for the official conference site, supporting 5,000+ visits.

ADDITIONAL

- **Certifications:** Google – IT Support; Deloitte – Data Analyst (Job Simulation)
- **Interests:** Chess, Swimming, Sketching, Badminton